

13.Car Price Prediction Model using Python

Aim

To implement a **Car Price Prediction Model** in Python using a machine learning algorithm and predict the price of a car based on its features.

Algorithm

1. Import the required libraries.
2. Create a dataset containing car features and prices.
3. Separate the input features and target price.
4. Split the dataset into training and testing data.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict the car prices for the test data.
8. Calculate the accuracy using R^2 score.
9. Predict the price of a new car.

PROGRAM:

```
from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import r2_score
```

```
X = [
    [2015, 50000, 1200],
    [2016, 45000, 1300],
    [2017, 40000, 1400],
    [2018, 35000, 1500],
    [2019, 30000, 1600],
    [2020, 25000, 1700],
    [2021, 20000, 1800],
    [2022, 15000, 1900],
```

```
[2023, 10000, 2000],  
[2024, 5000, 2100]  
]  
  
y = [  
    500000,  
    550000,  
    600000,  
    650000,  
    700000,  
    750000,  
    800000,  
    850000,  
    900000,  
    950000  
]  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
model = LinearRegression()  
model.fit(X_train, y_train)  
y_pred = model.predict(X_test)  
accuracy = r2_score(y_test, y_pred)
```

```
print("Actual Prices:", y_test)

print("Predicted Prices:", y_pred)

print("R2 Score:", accuracy)

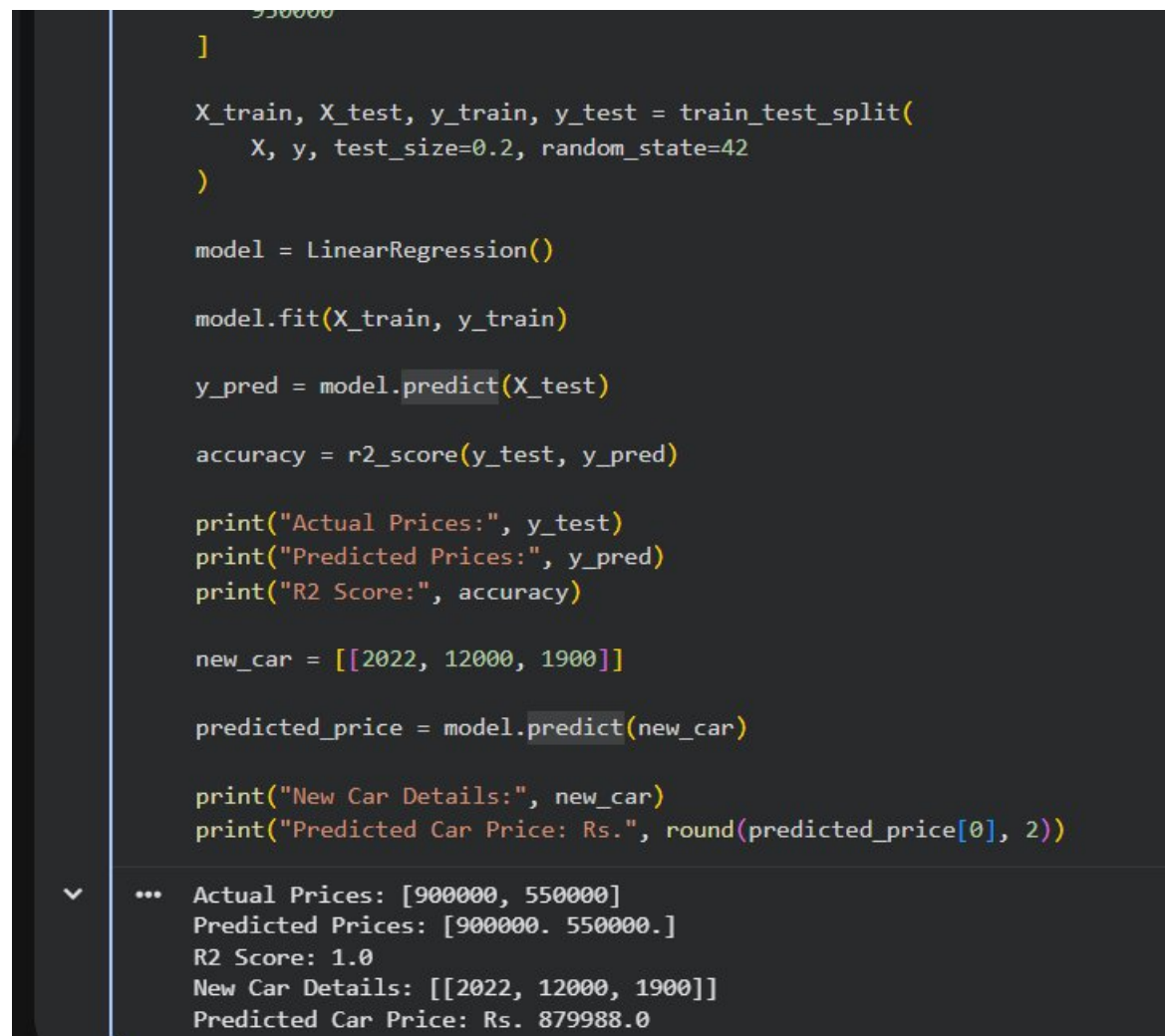
new_car = [[2022, 12000, 1900]]

predicted_price = model.predict(new_car)

print("New Car Details:", new_car)

print("Predicted Car Price: Rs.", round(predicted_price[0], 2))
```

OUTPUT:



```
    ]

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    model = LinearRegression()

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy = r2_score(y_test, y_pred)

    print("Actual Prices:", y_test)
    print("Predicted Prices:", y_pred)
    print("R2 Score:", accuracy)

    new_car = [[2022, 12000, 1900]]

    predicted_price = model.predict(new_car)

    print("New Car Details:", new_car)
    print("Predicted Car Price: Rs.", round(predicted_price[0], 2))
```

▼ ... Actual Prices: [900000, 550000]
Predicted Prices: [900000. 550000.]
R2 Score: 1.0
New Car Details: [[2022, 12000, 1900]]
Predicted Car Price: Rs. 879988.0

RESULT:

Thus, the **Car Price Prediction Model** was successfully implemented using **Linear Regression** in Python, and the price of a new car was successfully predicted.

